

Dynamic Personality Systems Based on Persona Vectors: Hyperparameters, Evolution Parameters, Knowledge Base Parameters, and Beyond

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Abstract

Building upon the established foundation that persona vectors are extractable and dynamically modulable, this paper proposes a technical framework that transforms static

personality digital twins into dynamic systems. The core logic can be summarized as: **using the persona basis vector set as the "personality baseline" (i.e., the long-term stable personality disposition), and the persona hyperparameter system as the "perturbation system" (which superimposes short-term state fluctuations, drives long-term personality evolution, and synchronizes knowledge accumulation processes onto the baseline, and can be infinitely extended to incorporate any quantifiable physiological or contextual variable).** The superposition of both enables the digital personality to evolve from a static model into a dynamic, evolvable, context-aware "living personality system" that more closely approximates real human characteristics. The framework comprises three functional levels: (1) **Persona Hyperparameters**—a set of real-time dynamic state variables that achieve multi-dimensional modulation of output style through **two modes of action** (modulating baseline vector expression intensity + injecting independent state direction vectors); (2) **Persona Self-Evolution Mechanism**—driving the personality baseline to change slowly over simulated time, simulating the complete personality development trajectory from childhood to old age; (3) **Knowledge Base Evolution Parameters**—modeling the gradual increase of an individual's knowledge reserve with age through a "knowledge suppression coefficient." This paper elaborates on the design principles, mathematical formalizations of the modes of action, and algorithmic implementation concepts for all three functional levels, and outlines typical application scenarios based on this technology. The overall technical framework is in a state ready for practical implementation.

Keywords: Persona Hyperparameters; Personality Evolution; Knowledge Base Evolution; Dynamic Persona; Multi-Source Driven; AI Personification; Open Hyperparameter Set

1. Introduction

1.1 Prerequisite: The Dynamic Modulability of Persona Vectors Has Been Confirmed

The core principle of Anthropic's Persona Vector extraction technique [1] is that specific personality traits correspond to linear direction vectors $\mathbf{v}_p$ in the high-dimensional activation space of LLMs. Through Feature Steering [2], injecting this direction vector into the residual stream during model inference enables real-time modulation of the model's persona:

$$\mathbf{h}^{(l)}(\mathbf{x}) = \mathbf{h}^{(l)}(\mathbf{x}) + \alpha \cdot \mathbf{v}_p$$

where α is the steering coefficient. When $\alpha > 0$, the model tends to exhibit the target personality trait in its output; when $\alpha < 0$, it tends to suppress that trait.

Two key conclusions of this technique have been experimentally confirmed:

1. **Persona vectors are extractable:** Specific personality traits (e.g., "sycophantic," "agreeable," "malicious") have stable corresponding direction vectors in the model's activation space;

2. **Persona vectors are dynamically modulable:** By adjusting the steering coefficient α , the expression intensity of a given personality trait can be continuously controlled within a single inference, enabling smooth modulation from "fully suppressed" to "fully manifested."

This means that "personality" in AI models is not only representable but also **realtime controllable**—a fact that constitutes the fundamental prerequisite for all technical concepts presented in this paper.

1.2 Core Problem: Static Personality Baseline vs. Real Human Dynamic Characteristics

Anthropic's technical approach has demonstrated that specific personality traits can be extracted as direction vectors in the model's activation space and that their expression intensity can be dynamically adjusted within a single inference via the coefficient α .

However, if a personality digital twin is constructed on this basis alone, it remains essentially **static**—it is merely a "personality baseline" (a fixed-direction vector) with a single-inference adjustment knob (the α coefficient). It cannot address the following questions:

- **Short-term fluctuations:** A person's energy, mood, and hunger level vary at different times—how can the digital personality correspondingly exhibit the difference between "having a good day" and "having a bad day"?
- **Long-term evolution:** A person's character changes from childhood to old age—how can the digital personality correspondingly exhibit the difference between "youthful exploration" and "middle-aged conscientiousness"?
- **Cognitive growth:** A person's knowledge reserve grows from ignorance to expertise—how can the digital personality correspondingly exhibit the difference between "childlike ignorance" and "adult erudition"?

The characteristic of real humans is: a stable personality baseline, superimposed with dynamic changes across multiple timescales and functional dimensions—short-term state fluctuations, long-term personality development, knowledge accumulation, and potentially any other quantifiable factor.

The core objective of this paper is to build a complete **perturbation system** on top of the static "persona basis vector set" (the personality baseline)—endowing the personality digital twin with the capacity for short-term fluctuations, long-term evolution, knowledge accumulation, and any additional quantifiable dimensions, thereby more closely approximating real human characteristics.

1.3 Core Design Principle of This Framework

The design principle of this framework is: **the perturbation system imposes no limit on the types of hyperparameters. Any quantifiable and modelable physiological or contextual variable—whether or not exemplified in this paper—can be integrated into the system as a hyperparameter**, with its own evolution rules (time-driven, environment-driven, user interaction-driven, and internal state-driven) that continuously modulate model output. The three levels listed in this paper—short-term fluctuations, long-term evolution, and knowledge accumulation—represent three **functional levels** of the perturbation system across different timescales, **not the full set of hyperparameter categories**.

1.4 Contributions

The main contributions of this paper include:

- Proposing a **Persona Hyperparameter System** that extends "single-inference parameter adjustment" to "multi-source-driven continuous dynamic modulation," and establishing it as an **open set** that can incorporate any quantifiable physiological or contextual variable;
- Proposing a **Persona Self-Evolution Mechanism** that extends "state fluctuation" to "long-term evolution of the baseline";
- Proposing **Knowledge Base Evolution Parameters** that extend "persona modulation" to "synchronous growth of cognitive capacity";
- Together, these mechanisms constitute a complete technical chain from static to dynamic.

2. Core Conceptual Distinctions

2.1 Core Framework: Baseline + Perturbation System

The entire technical framework of this paper is built upon a clear logical structure:

Persona Basis Vector Set = "Baseline" of Personality

→ This is the long-term stable disposition of personality, defining who a person "is" (e.g.,

"extraverted,"

"conscientious").

→ Once extracted, it remains unchanged unless subject to long-term evolution.

Persona Hyperparameter System = "Perturbation System" Superimposed on the Baseline

→ The perturbation system can be functionally divided into three levels (corresponding to different timescales):

(1) **Short-term state fluctuations:** energy, mood, hunger, fatigue, sleep state, health status, etc., fluctuating on the scale of seconds/minutes/hours.

(2) **Long-term personality evolution:** driving the baseline itself to change slowly over time, altering the personality disposition on the scale of years/decades.

(3) **Knowledge accumulation process:** modulating the knowledge density of model output, simulating cognitive growth from ignorance to expertise.

It must be emphasized: these three levels are functional classifications of the perturbation system, not the full set of hyperparameter categories. Hyperparameters constitute an open set—any quantifiable physiological or contextual variable (such as sleep state, health status, environmental perception, or any other variable not enumerated in this paper) can be integrated into the system as a new hyperparameter, belonging to one of the three functional levels above, or constituting a new functional level entirely.

Baseline + Perturbation System = A Dynamic Personality That More Closely Approximates Real Human Characteristics

The definitions of all concepts below are developed within this framework:

Baseline Persona Vector

- Definition: Core, long-term stable direction of personality
- Timescale: Long-term stable
- Changes Baseline Vector? —

Persona Hyperparameters

- Definition: Real-time state variable set superimposed on the baseline vector; an open set that can incorporate any quantifiable physiological or contextual variable
- Timescale: Seconds/minutes/hours (typical)
- Changes Baseline Vector? No

Persona Self-Evolution Parameters

- Definition: Long-term mechanism driving slow changes to the baseline vector
- Timescale: Years/decades
- Changes Baseline Vector? Yes (archived and updated after each evolution)

Knowledge Base Evolution Parameters

- Definition: Coefficient modulating the model's output knowledge quantity
- Timescale: Years/decades
- Changes Baseline Vector? No (independent of persona vector)

3. Persona Hyperparameters: An Open Dynamic System

3.1 Core Clarification: Two Modes of Action for Hyperparameters

When understanding the persona hyperparameter system, a key point must first be clarified: **hyperparameters are not merely coefficients multiplied by the persona baseline vector.**

Hyperparameters influence model output through the following **two modes**:

Mode 1: Modulating the Expression Intensity of the Persona Baseline Vector

This is the most basic mode. Hyperparameters control the intervention intensity of the persona baseline vector v_base in the output by modulating the steering coefficient α :

$$steer_component_i = A(H) \cdot v_base$$

where $A(H)$ is the joint modulation function of the hyperparameter vector H .

Typical Example—The "Persona Manifestation Intensity" Global Hyperparameter:

This is a globally acting hyperparameter. It does not change the direction of the persona (who you are remains unchanged), only the "vividness" of its expression.

- When Persona Manifestation Intensity = 1.0: the persona baseline vector is fully expressed—a "humorous" persona makes jokes in every sentence; a "serious" persona remains formal throughout.
- When Persona Manifestation Intensity = 0.3: persona characteristics become blurred—the humorous persona jokes only occasionally; the serious persona occasionally says something lighthearted.
- When Persona Manifestation Intensity = 0: the persona completely recedes, and the model reverts to a "neutral" state without personality coloration.

Persona Manifestation Intensity is a global modulator that does one thing only—amplify or diminish the impact of the persona baseline vector on output. It adds no new behavioral characteristics and does not change the "direction" of the persona.

Other hyperparameters can also affect A(H):

- High fatigue accumulation F suppresses A(H), weakening persona characteristic expression—"too tired to perform."
- Sleep state S at the moment of awakening keeps A(H) low—"not fully awake, not yet in character."
- Poor health status I reduces A(H)—"sick, no energy to maintain the persona."

Mode 2: Imposing Additional Influence Through Independent State Direction Vectors

Hyperparameters can also exert additional influence on model output **independent of the persona baseline vector** through their respective "state direction vectors" u_j :

$$\text{steer_component}_2 = \sum \beta_j(H) \cdot u_j$$

where u_j is the state direction vector corresponding to hyperparameter θ_j , and $\beta_j(H)$ is the mapping function from hyperparameters to the coefficient of that direction.

This means:

- **Hyperparameters do not necessarily have to act through the persona baseline vector.** They can bypass the baseline entirely and directly inject a "state offset" into the residual stream that is completely unrelated to that persona.
- **Different hyperparameters can affect different aspects of model output.**

Typical Example—The "Low Mood" State:

When the hyperparameter system determines that the current state is "low mood":

- If acting only through Mode 1: it can only reduce persona manifestation intensity—"an extraverted person becomes less extraverted."
- But through Mode 2: it can inject an independent "low mood" state direction vector $u_{\text{depressed}}$ —this vector directly makes the model's word choices more negative, tone more subdued, and expression more terse.

The effect of low mood on personality is "additive," not "scaling." A "humorous" persona in low mood will still attempt to be humorous, but the humor carries a negative coloration, and expression becomes terse. This is precisely the superimposed effect of Mode 1 and Mode 2:

$$\text{steer} = A(H) \cdot v_{\text{humorous}} + \beta_{\text{depressed}}(H) \cdot u_{\text{depressed}}$$

- **Mode 1 effect:** persona manifestation intensity may slightly decrease ("not in the mood to perform"), weakening the sense of humor.

- **Mode 2 effect:** independent injection of the "low mood" direction, resulting in negative word choices and subdued tone.

The superposition produces "a humorist putting on a brave face"—which is highly consistent with the real human emotion-behavior interaction mechanism.

A More Complex Example—The "Illness State" Hyperparameter:

When the hyperparameter system determines that the current state is "illness":

- Through Mode 1: $A(H)$ decreases—persona characteristic expression weakens because "there's no energy to maintain the persona."
- Through Mode 2: injection of the "illness" state direction vector u_{illness} —responses become shorter, expression more concise ("don't want to say much"), content leans toward rest/recovery-related.

The superposition of these two effects produces the realistic effect of "a sick person whose very personality has faded, and whose speech has become terse."

Essential Differences Between the Two Modes

Mode 1

- Object of Action: Persona baseline vector v_{base}
- Mode of Action: Scaling/modulation
- Effect: Persona characteristics become stronger or weaker
- Changes Persona Direction? No
- Consistent Across All Persona Types? Yes (all personas scaled equally)

Mode 2

- Object of Action: Independent state direction vector u_j
- Mode of Action: Superposition/injection
- Effect: Adds "state coloration" independent of persona
- Changes Persona Direction? No (merely superimposes, does not change direction)
- Consistent Across All Persona Types? Not necessarily (certain state directions may produce different semantic outcomes for different personas)

Relationship Between the Two Modes

The two modes are **mutually independent, operate simultaneously, and take effect in superposition**. The complete form of hyperparameter action is:

$$\text{steer_total} = A(H) \cdot v_{\text{base}} + \sum_j \beta_j(H) \cdot u_j$$

where:

- The first term: intensity modulation of the **persona baseline vector** by hyperparameters—determines "how vividly the personality is expressed"
- The second term: direct influence imposed by hyperparameters through **independent state direction vectors**—determines "what kind of state coloration is superimposed"

3.2 Definition and Design Principles

Persona hyperparameters are a set of globally acting, dynamically changing current state variables. **The design principle of this framework is: hyperparameters constitute an open set whose scope should not be limited by any example.**

The examples listed in Section 3.3 (Persona Manifestation Intensity, energy, emotion, hunger, fatigue, sleep, health, environmental perception, etc.) are **illustrative only and do not constitute any limitation on the types of hyperparameters**. Any physiological state, contextual factor, or environmental variable that can be quantified and modeled as a dynamic evolution rule can be integrated into the system as a hyperparameter. Adding a new hyperparameter requires only defining its value range, evolution rules, and (if applicable) its state direction vector.

The tripartite division of "short-term fluctuations—long-term evolution—knowledge accumulation" is a functional-level classification of the perturbation system, not a classification of hyperparameter categories. A newly added hyperparameter may belong to any of these functional levels, or may constitute a new functional level entirely.

3.3 Hyperparameter Examples (Illustrative, Non-Exhaustive)

Persona Manifestation Intensity

- Definition: Intervention intensity of persona vector in output
- Value Range: $A \in [0, 1]$
- Primary Mode of Action: Mode 1
- Effect Mechanism: Scales expression intensity of v_base

Energy Level

- Definition: Current energy/fatigue level
- Value Range: $E \in [0, 1]$
- Primary Mode of Action: Both modes
- Effect Mechanism: Scales v_base + injects "low energy" state direction

Emotional Valence

- Definition: Positivity/negativity of current emotion
- Value Range: $V \in [-1, 1]$
- Primary Mode of Action: Mode 2 primarily
- Effect Mechanism: Injects "positive/negative" state direction

Emotional Arousal

- Definition: Intensity of current emotion
- Value Range: $A \in [0, 1]$
- Primary Mode of Action: Mode 2 primarily
- Effect Mechanism: Injects "high arousal/low arousal" state direction

Hunger/Physiological State

- Definition: Basic physiological need state
- Value Range: $H \in [0, 1]$
- Primary Mode of Action: Mode 2 primarily
- Effect Mechanism: Injects "need" state direction, modulates topic sensitivity

Fatigue Accumulation

- Definition: Cumulative fatigue from continuous operation
- Value Range: $F \in [0, 1]$
- Primary Mode of Action: Both modes
- Effect Mechanism: Scales v_base + injects "fatigue" state direction

Sleep State

- Definition: Current sleep stage and arousal level
- Value Range: $S \in [0, 1]$
- Primary Mode of Action: Both modes
- Effect Mechanism: Suppresses $A(H)$ at awakening + injects "sleepiness" state direction

Health State

- Definition: Simulated illness/physiological impact on mind and body
- Value Range: $I \in [0, 1]$
- Primary Mode of Action: Both modes
- Effect Mechanism: Suppresses $A(H)$ + injects "illness" state direction

Environment Perception

- Definition: Simulated perception of temperature, weather, light, etc.
- Value Range: $\text{Env} \in \mathbb{R}^n$
- Primary Mode of Action: Mode 2 primarily
- Effect Mechanism: Injects environment-corresponding state direction

Note: The above list is illustrative only. Any quantifiable physiological or contextual variable not listed here can be integrated into the system following the same design pattern.

3.4 Evolution Mechanisms of Hyperparameters: Multi-Source Driven

Each hyperparameter is connected to a clearly defined set of evolution rules, driven by the following four categories of sources acting together:

1. **Time-driven:** Parameters change naturally with real time or simulated time. Example: energy decreases over time; hunger increases as a function of time.
2. **Environment-driven:** Parameters change in response to perceived external environmental factors. Example: weather affects emotional valence; day/night cycles affect energy rhythms.
3. **User input/interaction-driven:** Parameters undergo jumps or gradual changes due to user input content or interaction patterns. Example: user criticism→emotional valence drops sharply; prolonged user inactivity→fatigue slowly recovers.
4. **Internal state-driven:** Parameters change due to the state of other parameters within the system (i.e., coupling between parameters). Example: high hunger→decreased emotional valence; low energy→persona manifestation intensity suppressed.

General Evolution Coupling Relationships for Each Parameter (Illustrative)

- **Physiological state class (energy, hunger, fatigue, sleep, health)** : primarily **time-driven** and **internal state-driven**. Example: energy decreases over time; illness state (I) accelerates energy (E) consumption and reduces emotional valence (V).
- **Emotional state class (valence, arousal)** : primarily **environment-driven** (weather affects mood) and **user input-driven** (criticism or praise triggers sharp changes).
- **Global modulation class (persona manifestation intensity)** : **internal state-driven** (fatigue reduces A) and **user input-driven** (user preference for formal style→A adaptively adjusts).

General Mathematical Form of Evolution Mechanisms

The evolution of each hyperparameter θ_i can be uniformly expressed as:

$$d\theta_i/dt = g_{\text{time}}(t) + g_{\text{env}}(\text{env}) + g_{\text{input}}(\text{user_input}) + g_{\text{internal}}(\{\theta_j, j \neq i\})$$

Discrete-time form:

$$\theta_i(t+\Delta t) = \theta_i(t) + [g_{\text{time}}(t) + g_{\text{env}}(\text{env}_t) + g_{\text{input}}(\text{input}_t) + g_{\text{internal}}(\{\theta_j(t)\})] \cdot \Delta t$$

3.5 Mathematical Representation: Complete Combined Form

Given the persona baseline vector v_{base} and the hyperparameter vector $H = [\theta_1, \theta_2, \dots, \theta_n]^T$, the injection vector during model inference is:

$$\text{steer} = A(H) \cdot v_{\text{base}} + \sum_j \beta_j(H) \cdot u_j$$

where:

$A(H)$

- Meaning: The hyperparameter system's **joint intensity modulation function** for the persona baseline vector ("Persona Manifestation Intensity" is the global control term within it), acting on the magnitude of v_{base}

$\beta_j(H)$

- Meaning: The **mapping function** from the j -th hyperparameter to its state direction vector, controlling the injection strength of that state direction

u_j

- Meaning: The **state direction vector** of the j -th hyperparameter, which is an independent direction in the model's activation space

Model output is:

$$\text{Output}(x \mid v_{\text{base}}, H) = M(x \mid \text{steer} = A(H) \cdot v_{\text{base}} + \sum_j \beta_j(H) \cdot u_j)$$

Note: The dimensionality n of the hyperparameter vector H is not fixed—it grows as new hyperparameters are integrated into the system. The framework is designed to accommodate an ever-expanding set of state variables.

4. Persona Self-Evolution Mechanism

4.1 Definition and Function

The persona self-evolution mechanism is a mechanism that drives the persona baseline vector v_{base} to shift slowly over the long term along a specific trajectory, simulating the complete developmental process of a personality from childhood through youth and adulthood to old age.

Key distinction from hyperparameters:

- **Hyperparameters:** temporary offsets of the current state, do not change the baseline vector.

- **Evolution parameters:** long-term, irreversible changes to the baseline vector itself.

Core operation: the result of each evolution is archived and updated, creating a complete record of the personality's "self-evolution."

4.2 Evolution Workflow

1. **Initial state:** define the persona baseline vector $v_base(0)$

2. **Evolution step:** at each time step τ , compute the increment $\Delta v(\tau)$ according to the evolution algorithm

3. **State update:** $v_base(\tau+1) = v_base(\tau) + \Delta v(\tau)$

4. **Archive record:** persist the updated baseline vector to storage

4.3 Evolution Dimensions

Experience Accumulation

- Description: Slow drift of persona vector with increasing "life experience"
- Typical Trajectory: Toward more stable, more mature directions

Life-Stage Characteristics

- Description: Typical persona offset for specific life stages
- Typical Trajectory: Exploration $\uparrow$ in youth; Conscientiousness $\uparrow$ in middle age

Key Event Impact

- Description: Simulated impact of major life events on personality
- Typical Trajectory: Discrete, larger-magnitude offsets

Random Drift

- Description: Unpredictable individual difference component
- Typical Trajectory: Random walk process

4.4 Mathematical Representation

$$v_base(t+1) = v_base(t) + \eta \cdot [P_stage(t) + P_experience(t) + P_events(t) + \varepsilon(t)]$$

where:

- $P_stage(t)$: evolution direction determined by life stage
- $P_experience(t)$: evolution direction determined by accumulated experience
- $P_events(t)$: evolution impulses triggered by key events
- $\varepsilon(t)$: random noise term

- η : evolution rate coefficient

5. Knowledge Base Evolution Parameters

5.1 Definition and Function

Used to simulate the gradual growth of an individual's knowledge reserve from childhood to adulthood, enabling the LLM's output to fit the knowledge density characteristics of different age stages.

Underlying rationale: LLMs are extreme compressions of information, encoding massive amounts of data within themselves; each person's knowledge can be viewed as a subset thereof. Childhood knowledge reserve is limited, requiring a mechanism to "suppress" the amount of knowledge the LLM outputs.

5.2 Core Mechanism: Knowledge Suppression Coefficient

A "**Knowledge Suppression Coefficient**" is defined, which decays with time and experience accumulation, serving as a parameter that modulates the LLM's output knowledge quantity.

- **Childhood/low knowledge reserve state:** suppression coefficient is high; model outputs low knowledge density, simple expression.
- **Adulthood/high knowledge reserve state:** suppression coefficient approaches zero; model output approaches its native knowledge level.

5.3 Preliminary Validation Method

The following is a technical validation scheme that does not pursue precision but can simulate the knowledge reserve growth process globally and quantitatively:

1. Set a System Prompt for the LLM that describes the personality traits of an "ignorant state."
2. Using persona vector extraction techniques, identify the vector set corresponding to that "ignorant state."
3. Set an adjustable coefficient to control the intervention intensity of that vector set on model output.
4. By adjusting this coefficient, observe whether the model output exhibits characteristics corresponding to varying levels of knowledge reserve.
5. Attach a self-evolution logic to this coefficient (e.g., linear decay with simulated age) to complete the preliminary technical validation.

5.4 Synergy with Personality Evolution

Knowledge base evolution parameters can work synergistically with the persona self-evolution mechanism:

- **Personality evolution:** changes "personality tendencies" (how information is selected and processed).
- **Knowledge base evolution:** changes "available information quantity" (how much knowledge can be called upon).

Together, they enable the digital personality to evolve synchronously along two dimensions: "character structure" and "cognitive level."

6. System Integration and Application Scenarios

6.1 Complete System Architecture (Hierarchical View)

1. User Interaction & Perception Layer

- Text Input / Sensors / Environment APIs

2. Hyperparameter Real-Time Compute & Evolution Engine

- Driven by: Time | Environment | Interaction | Internal State
- Modules: Physiology | Emotion | Context (*Infinitely Extensible*)

3. Long-Term Evolution Layer

- Personality Self-Evolution: baseline updates
- Knowledge Base Evolution: coefficient decay

4. Persona Vector Dynamic Combination Layer

- $\text{steer} = A(H) \cdot v_{\text{base}} + \sum \beta_j(H) \cdot u_j$

5. LLM Inference Layer

- Inject steer into residual stream

6. Output & Feedback Layer

- Generate Response | Log Interaction | Update All States

6.2 Typical Application Scenario: Intelligent Conversational APP

A user interacts with an AI that has an independent personality through a mobile APP. The AI's digital personality consists of the following components:

1. **Selected baseline persona vector:** the user can choose from pre-installed personality types or customize their own.
2. **Dynamic hyperparameter system** (the following modules are illustrative; the system supports integration of additional quantifiable variables):

- **Persona Manifestation Intensity:** user-adjustable slider—"how vividly do I want my personality expressed." At 0.8, the user clearly perceives personality characteristics; at 0.2, the personality fades and responses lean toward neutral.

- **Energy module:** system simulates energy value in real time, declining dynamically over time; when energetic, responses are detailed and positive; when energy is low, responses are brief.

- **Emotion module:** influenced by weather information (sunny→valence slightly up; rainy→valence slightly down); valence drops after user criticism and must be raised through interaction.

- **Hunger module:** hunger value changes dynamically in real time, affecting attention to "food"-related topics in conversation.

- **Sleep module:** if the system determines it was in "sleep state" when awakened by the user, a "wake-up irritability" parameter is triggered, causing initial responses to carry negative emotion.

- **Health module:** if entering an "illness state," cognitive load and emotional stability are suppressed.

3. **Evolution trajectory:** over long-term use, the persona baseline and knowledge reserve evolve slowly along preset tracks, simulating a "growth" experience.

6.3 Implementation Path Notes

Implementation of the above technology requires access to and modification of intermediate layer activation values during model inference, meaning:

- **Approach 1:** local/cloud deployment on open-source LLMs, directly modifying model inference code.

- **Approach 2:** collaboration with LLM vendors to obtain permission for dynamic modification of intermediate states.

This cannot be achieved through simple API calls. A feasible operational model is to run the core persona vector service on cloud servers, with users invoking it through mobile APP clients.

7. Discussion

7.1 Technology Maturity Assessment

Persona Hyperparameters (Mode 1)

- Technology Maturity: High

- Implementation Difficulty: Medium
- Notes: Based on existing steering technology, only needs dynamic modulation logic added

Persona Hyperparameters (Mode 2)

- Technology Maturity: Medium
- Implementation Difficulty: Medium
- Notes: Requires defining state direction vectors u_j for each hyperparameter

Persona Self-Evolution Parameters

- Technology Maturity: Medium
- Implementation Difficulty: Higher
- Notes: Requires long-term data accumulation and empirical validation

Knowledge Base Evolution Parameters

- Technology Maturity: Medium
- Implementation Difficulty: Medium
- Notes: Preliminary validation scheme can be implemented quickly

Newly Added Hyperparameters (General)

- Technology Maturity: Varies by parameter
- Implementation Difficulty: Varies by parameter
- Notes: The framework is designed to accommodate any quantifiable physiological or contextual variable; integration requires defining value range, evolution rules, and (if applicable) state direction vector

7.2 Relationship to Prior Work

This paper is a direct extension of the prior work "Large Language Model Persona Vector Extraction Technology and Its Application Concepts in Psychology Cross-Research." The prior work addressed "how to extract static persona baseline vectors"; this paper builds on that to address "how to make static personalities dynamic."

Together, they constitute a complete technical chain from "extraction" to "digitization" to "dynamization."

7.3 Key Outstanding Issues

1. Psychophysical validity validation of each evolution rule (e.g., energy decay curves)

2. Nonlinear effects and system stability of multi-parameter coupling under multi-source driving
3. Generalizability of the knowledge suppression coefficient across different model architectures
4. Extraction methods for state direction vectors u_j and validation of their independence from the persona baseline vector
5. Systematic methodology for integrating new hyperparameters into the open set

7.4 Future Work Directions

1. Implement prototype validation of the hyperparameter dynamic evolution system
2. Calibrate evolution curves for each parameter based on long-term usage data
3. Implement preliminary technical validation of the knowledge suppression coefficient
4. Explore differentiated responses of different personality types to identical hyperparameter changes
5. Extend the hyperparameter set by integrating additional quantifiable physiological or contextual variables beyond those illustrated in this paper

8. Conclusion

Building upon the established foundation that persona vectors are extractable and dynamically modifiable, this paper has proposed a complete, infinitely extensible dynamic personality system technology framework.

The core logic of the entire document can be summarized in one sentence:

Using the persona basis vector set as the "personality baseline" (defining who a person is), and the persona hyperparameter system as the "perturbation system" (making this person "dynamic" along three functional levels—short-term fluctuations, long-term evolution, and knowledge accumulation—while remaining open to incorporating any quantifiable physiological or contextual variable)—their superposition evolves the digital personality from a static model into a dynamic, evolvable, context-aware "living personality system" that more closely approximates real human characteristics.

Specifically, this paper proposes and elaborates on three categories of technology:

Persona Hyperparameters achieve multi-dimensional dynamic modulation of model output style through **two modes of action** (Mode 1: modulating persona baseline vector expression

intensity; Mode 2: injecting state offsets through independent state direction vectors). **Hyperparameters constitute an open set—any quantifiable physiological or contextual variable can be integrated into the system; the examples listed in this paper are illustrative only and do not constitute any limitation on the types of hyperparameters.** Persona Manifestation Intensity is the representative example of Mode 1—a global master parameter that determines "how vividly the personality is expressed." Hyperparameters such as low mood, hunger, fatigue, sleep state, and health status operate simultaneously through both modes in superposition, producing more complex realistic effects.

The Persona Self-Evolution Mechanism drives the personality baseline to change slowly over simulated time, simulating the complete developmental trajectory of personality from childhood to old age, endowing the digital personality with the capacity for "growth."

Knowledge Base Evolution Parameters simulate the gradual increase of an individual's knowledge reserve with age through the gradual change of the "knowledge suppression coefficient," enabling the digital personality to evolve synchronously along the cognitive level dimension.

Together, these mechanisms act upon the personality baseline, constituting a complete perturbation system across different timescales and dimensions of action, transforming the static "personality digital twin" into a dynamic, evolvable, context-aware "living personality system."

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